

Uni-Mol

训练结果分析报告

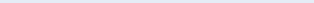
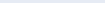
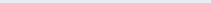
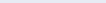
11111111

■ ■	■ ■	■ ■ ■ ■	■ ■ ■ ■
ESOL	■ ■ ■ ■ ■	Pearson r = 0.949	■ ■ ■
BBBP	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	AUC = 0.899	■ ■ ■
Tox21	■ ■ ■ ■ ■ ■ ■	AUC = 0.648	■ ■ ■
QM9DFT	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	Pearson r = 0.980	■ ■ ■

log mol/L

6 of 10

Model	MAE	MSE	R ²	Spearman r	Pearson r
Model 1	0.4692	0.4309	0.8992	0.9470	0.9487
Model 2	0.4711	0.4321	0.8978	0.9458	0.9475
Model 3	0.4723	0.4334	0.8965	0.9445	0.9462
Model 4	0.4735	0.4346	0.8952	0.9432	0.9449
Model 5	0.4747	0.4358	0.8939	0.9419	0.9436
Model 6	0.4759	0.4370	0.8926	0.9406	0.9423
Model 7	0.4771	0.4382	0.8913	0.9393	0.9410
Model 8	0.4783	0.4394	0.8900	0.9380	0.9397
Model 9	0.4795	0.4406	0.8887	0.9367	0.9384
Model 10	0.4807	0.4418	0.8874	0.9354	0.9371
Model 11	0.4819	0.4430	0.8861	0.9341	0.9358
Model 12	0.4831	0.4442	0.8848	0.9328	0.9345
Model 13	0.4843	0.4454	0.8835	0.9315	0.9332
Model 14	0.4855	0.4466	0.8822	0.9302	0.9319
Model 15	0.4867	0.4478	0.8809	0.9289	0.9306
Model 16	0.4879	0.4490	0.8796	0.9276	0.9293
Model 17	0.4891	0.4502	0.8783	0.9263	0.9280
Model 18	0.4903	0.4514	0.8770	0.9250	0.9267
Model 19	0.4915	0.4526	0.8757	0.9237	0.9254
Model 20	0.4927	0.4538	0.8744	0.9224	0.9241
Model 21	0.4939	0.4550	0.8731	0.9211	0.9228
Model 22	0.4951	0.4562	0.8718	0.9198	0.9215
Model 23	0.4963	0.4574	0.8705	0.9185	0.9202
Model 24	0.4975	0.4586	0.8692	0.9172	0.9189
Model 25	0.4987	0.4598	0.8679	0.9159	0.9176
Model 26	0.4999	0.4610	0.8666	0.9146	0.9163
Model 27	0.5011	0.4622	0.8653	0.9133	0.9150
Model 28	0.5023	0.4634	0.8640	0.9120	0.9137
Model 29	0.5035	0.4646	0.8627	0.9107	0.9124
Model 30	0.5047	0.4658	0.8614	0.9094	0.9111
Model 31	0.5059	0.4670	0.8601	0.9081	0.9098
Model 32	0.5071	0.4682	0.8588	0.9068	0.9085
Model 33	0.5083	0.4694	0.8575	0.9055	0.9072
Model 34	0.5095	0.4706	0.8562	0.9042	0.9059
Model 35	0.5107	0.4718	0.8549	0.9029	0.9046
Model 36	0.5119	0.4730	0.8536	0.9016	0.9033
Model 37	0.5131	0.4742	0.8523	0.9003	0.9020
Model 38	0.5143	0.4754	0.8510	0.8990	0.9007
Model 39	0.5155	0.4766	0.8497	0.8977	0.8994
Model 40	0.5167	0.4778	0.8484	0.8964	0.8981
Model 41	0.5179	0.4790	0.8471	0.8951	0.8968
Model 42	0.5191	0.4802	0.8458	0.8938	0.8955
Model 43	0.5203	0.4814	0.8445	0.8925	0.8942
Model 44	0.5215	0.4826	0.8432	0.8912	0.8929
Model 45	0.5227	0.4838	0.8419	0.8899	0.8916
Model 46	0.5239	0.4850	0.8406	0.8886	0.8903
Model 47	0.5251	0.4862	0.8393	0.8873	0.8890
Model 48	0.5263	0.4874	0.8380	0.8860	0.8877
Model 49	0.5275	0.4886	0.8367	0.8847	0.8864
Model 50	0.5287	0.4898	0.8354	0.8834	0.8851
Model 51	0.5299	0.4910	0.8341	0.8821	0.8838
Model 52	0.5311	0.4922	0.8328	0.8808	0.8825
Model 53	0.5323	0.4934	0.8315	0.8795	0.8812
Model 54	0.5335	0.4946	0.8302	0.8782	0.8799
Model 55	0.5347	0.4958	0.8289	0.8769	0.8786
Model 56	0.5359	0.4970	0.8276	0.8756	0.8773
Model 57	0.5371	0.4982	0.8263	0.8743	0.8760
Model 58	0.5383	0.4994	0.8250	0.8730	0.8747
Model 59	0.5395	0.5006	0.8237	0.8717	0.8734
Model 60	0.5407	0.5018			

ESOL  > 0.94 $R^2 = 0.90$ 
Uni-Mol  MSE/MAE  0.47  log 

BBB

Model		Model	Model	Model
AUC / AUROC	0.8985			
AUPRC	0.9737			
Accuracy	0.9056			
F1-Score	0.9452	F1		
Precision	0.9216			
Recall	0.9701			

MCC	0.6166		
Cohen's Kappa	0.6061		
Log Loss	0.3676		

BBBP AUC = 0.90 > 92% F1 = 0.9452 MCC Kappa ~0.61

3. Tox21

21 Nuclear receptor signaling Stress response pathways

AUC / AUROC	0.6475		
AUPRC	0.2377		

□ □ □ □ □ □ □ □ □ □ □ □ □ □

11/11/2019

- 11/11/11

- #### 4. QM9DFT

Page 10 of 10

11

5 of 10

Page 10 of 10

4 of 4

5

ESOL		
BBBP		MCC
Tox21		Focal Loss

-
1. Uni-Mol QM9DFT ESOL
 2. BBBP AUC > 0.85 MCC
 3. Tox21 AUC/MCC
 4. Tox21

2026 Uni-Mol/5